Exercício 3.28 do Livro Algoritmos

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a)

x_1	x_2	x_3	x_4	$(x_1 \vee \overline{x_2})$	$(\overline{x_1} \vee \overline{x_3})$	$(x_1 \vee x_2)$	$(\overline{x_3} \vee x_4)$	$(\overline{x_1} \vee x_4)$	Todas as Cláusulas
0	0	0	0	1	1	0	1	1	0
0	0	0	1	1	1	0	1	1	0
0	0	1	0	1	1	0	0	1	0
0	0	1	1	1	1	0	1	1	0
0	1	0	0	0	1	1	1	1	0
0	1	0	1	0	1	1	1	1	0
0	1	1	0	0	1	1	0	1	0
0	1	1	1	0	1	1	1	1	0
1	0	0	0	1	1	1	1	0	0
1	0	0	1	1	1	1	1	1	1
1	0	1	0	1	0	1	0	0	0
1	0	1	1	1	0	1	1	1	0
1	1	0	0	1	1	1	1	0	0
1	1	0	1	1	1	1	1	1	1
1	1	1	0	1	0	1	0	0	0
1	1	1	1	1	0	1	1	1	0

b)

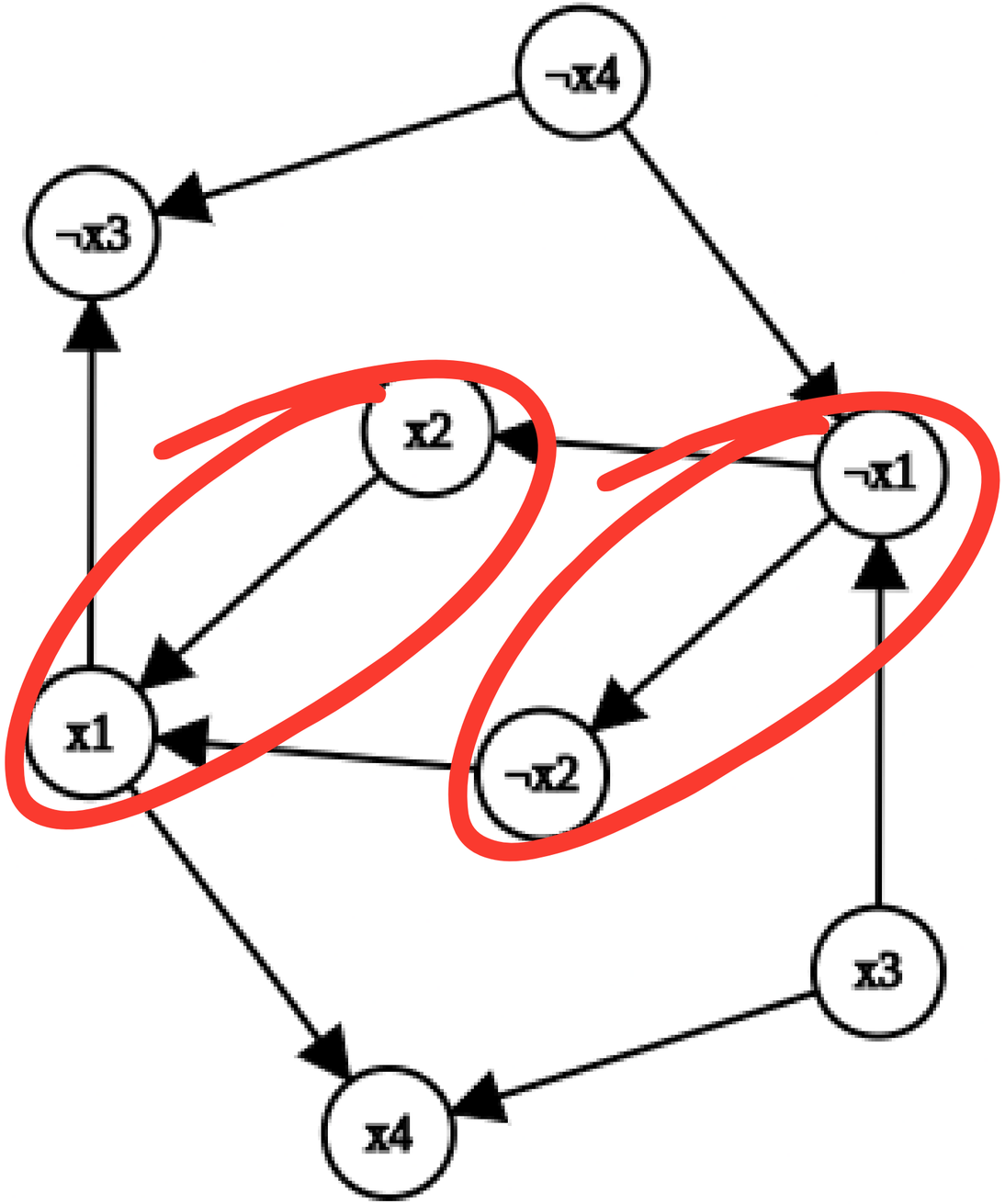
x_1	$(x_1 \vee x_1)$	$(x_1 \vee \overline{x_1})$	$(\overline{x_1} \vee \overline{x_1})$	Todas as Cláusulas
0	0	1	1	0
1	1	1	0	0

$(x1 \vee x1) \equiv x1 \quad (\neg x1 \vee \neg x1) \equiv \neg x1$

Instância Insatisfatível:

(x1 v x1)(¬x1 v ¬x1)(¬x1 v ¬x2)(x3 v x4)

c)

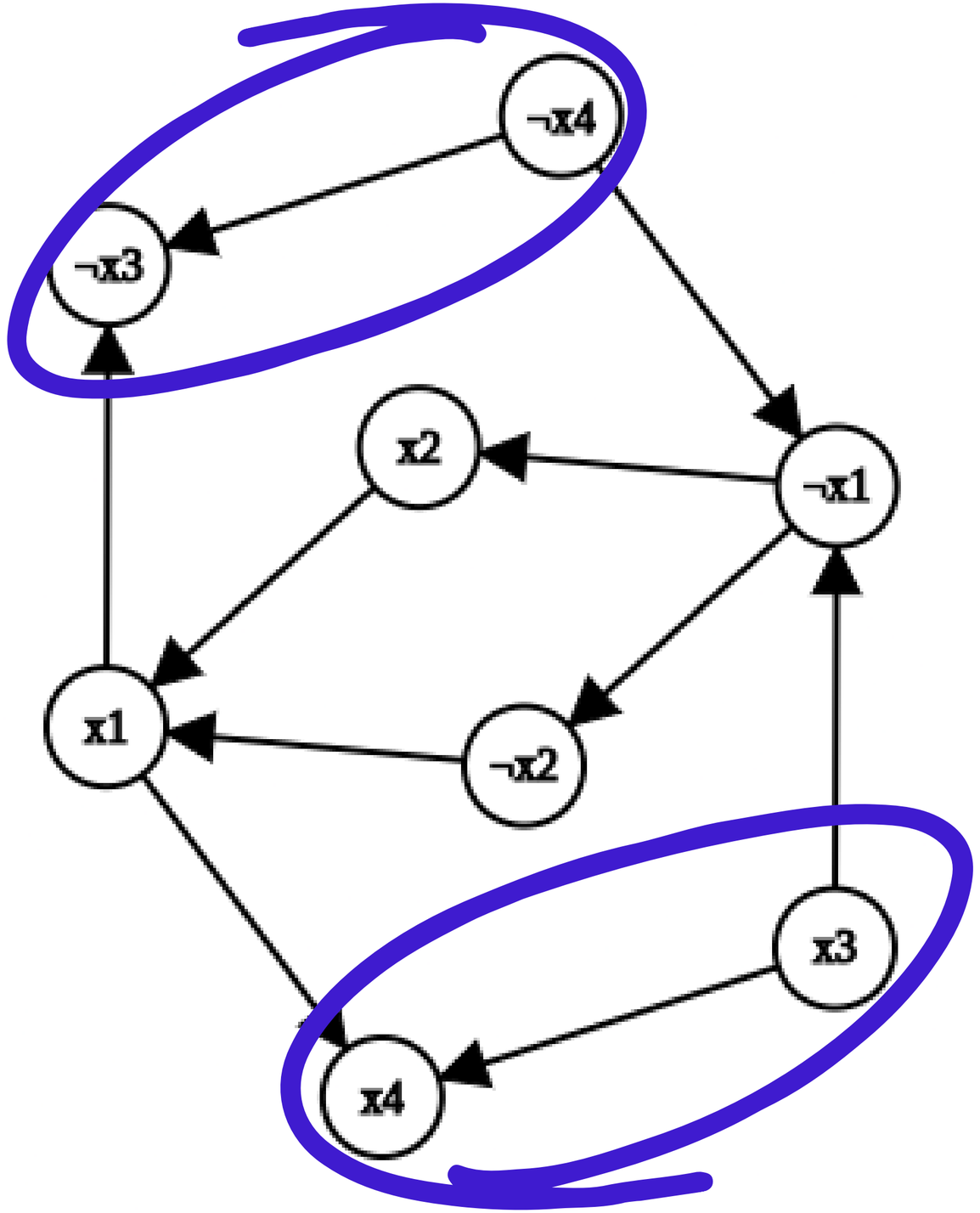


(x1 v ¬x2) (¬x3 v x4)

(x1 v x2) (¬x1 v x4)

(¬x1 v ¬x3)

c)

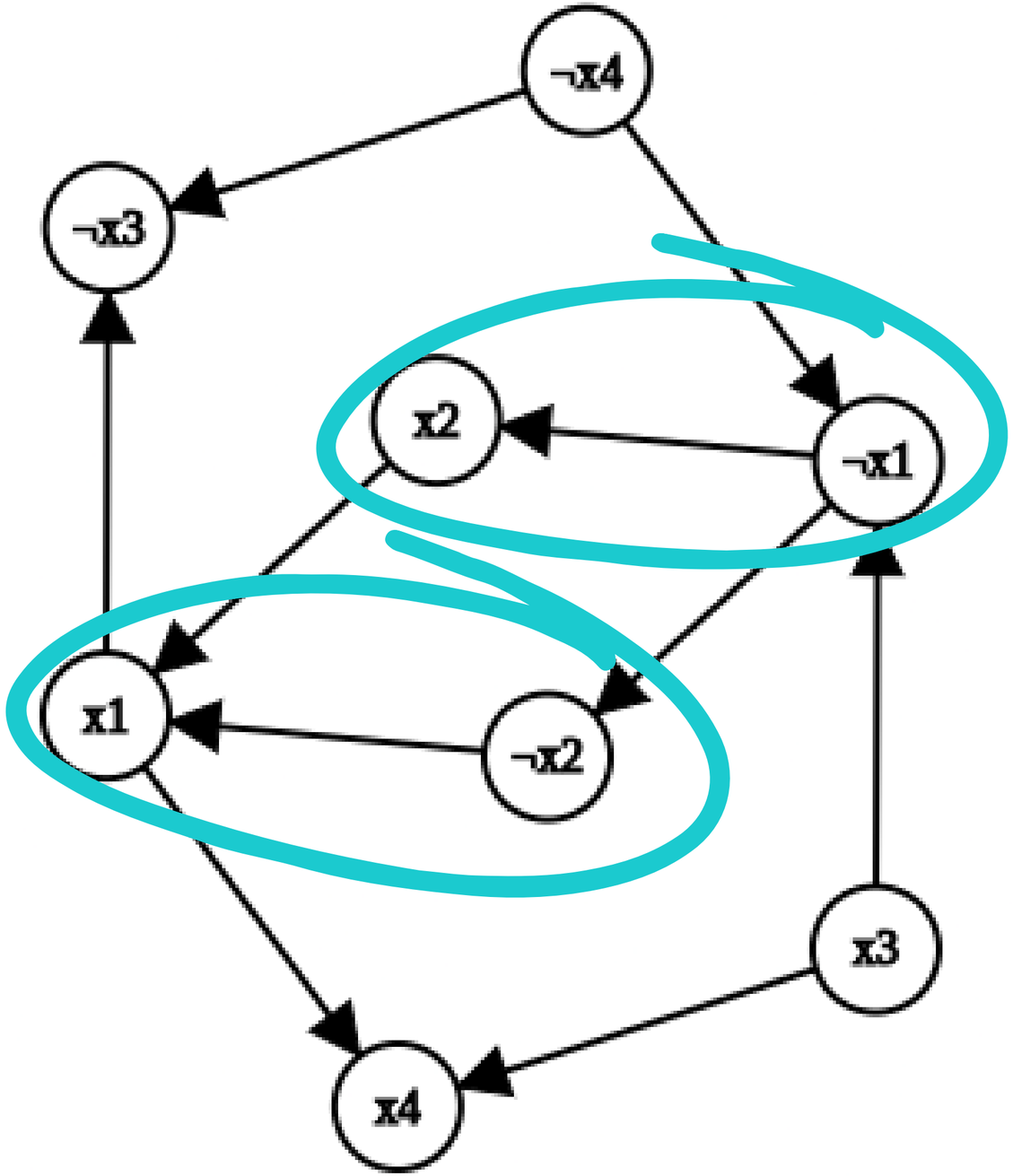


(x1 v ¬x2) (¬x3 v x4)

(x1 v x2) (¬x1 v x4)

(¬x1 v ¬x3)

c)

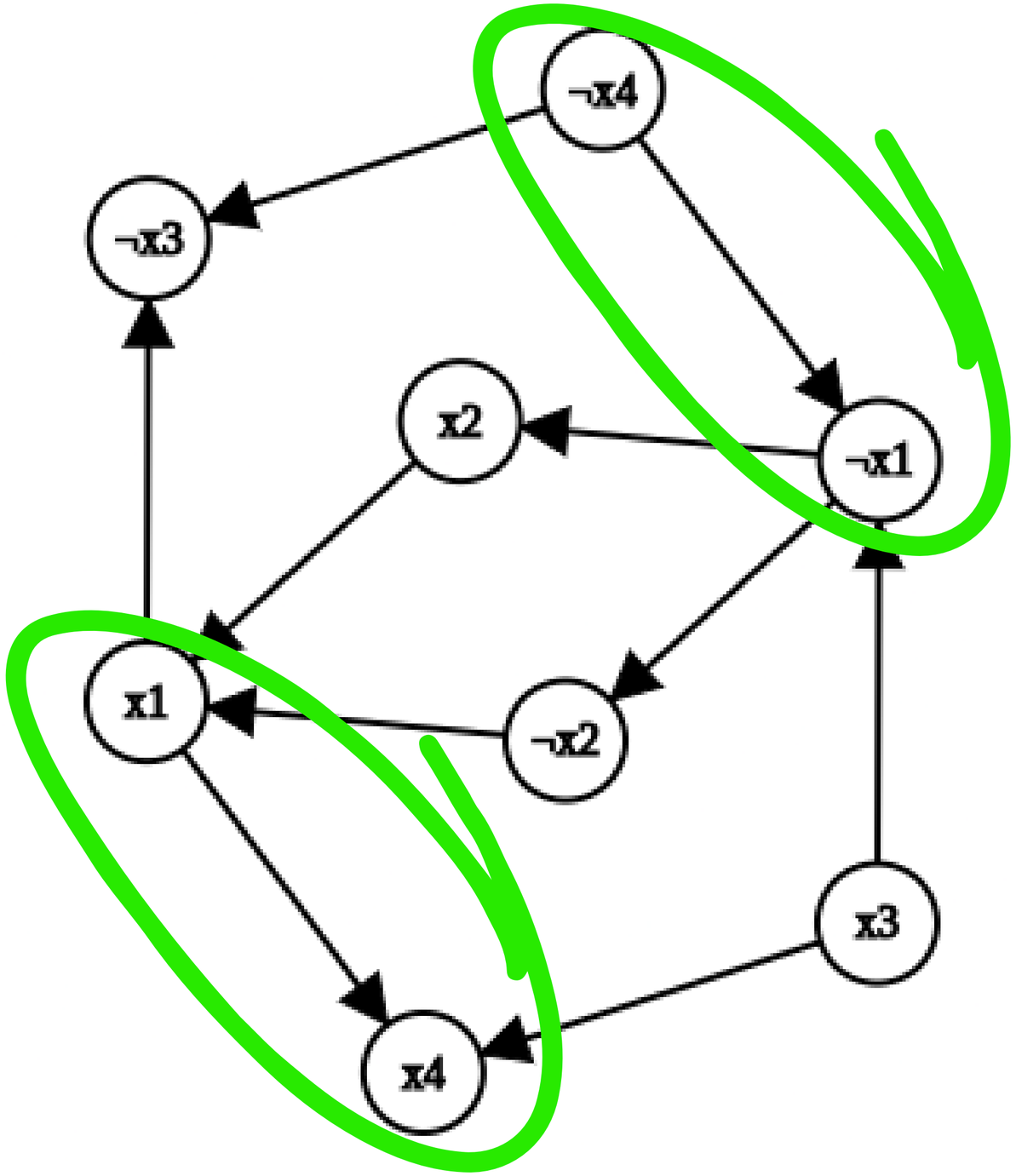


(x1 v ¬x2) (¬x3 v x4)

(x1 v x2) (¬x1 v x4)

(¬x1 v ¬x3)

c)

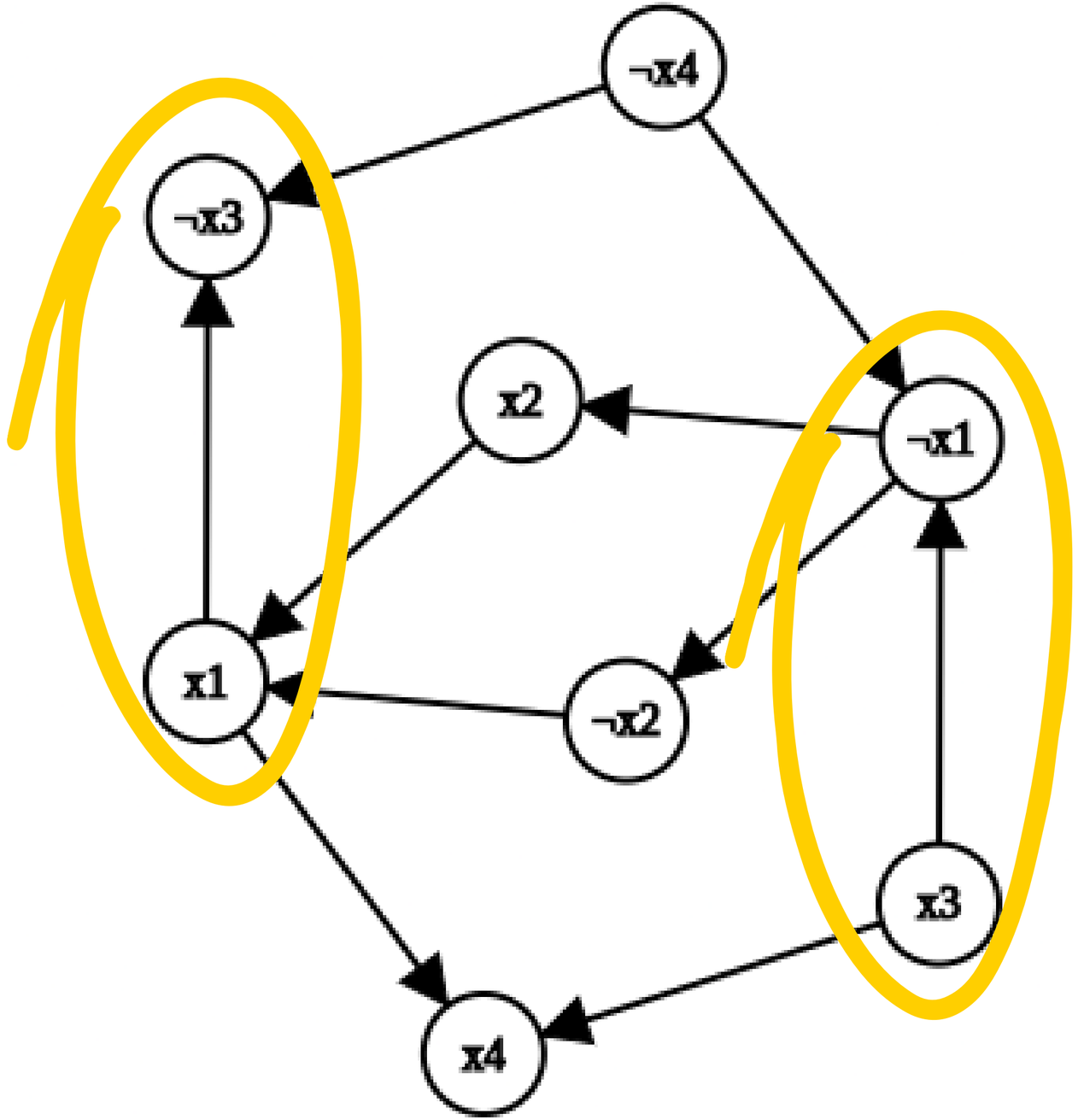


(x1 v ¬x2) (¬x3 v x4)

(x1 v x2) (¬x1 v x4)

(¬x1 v ¬x3)

c)

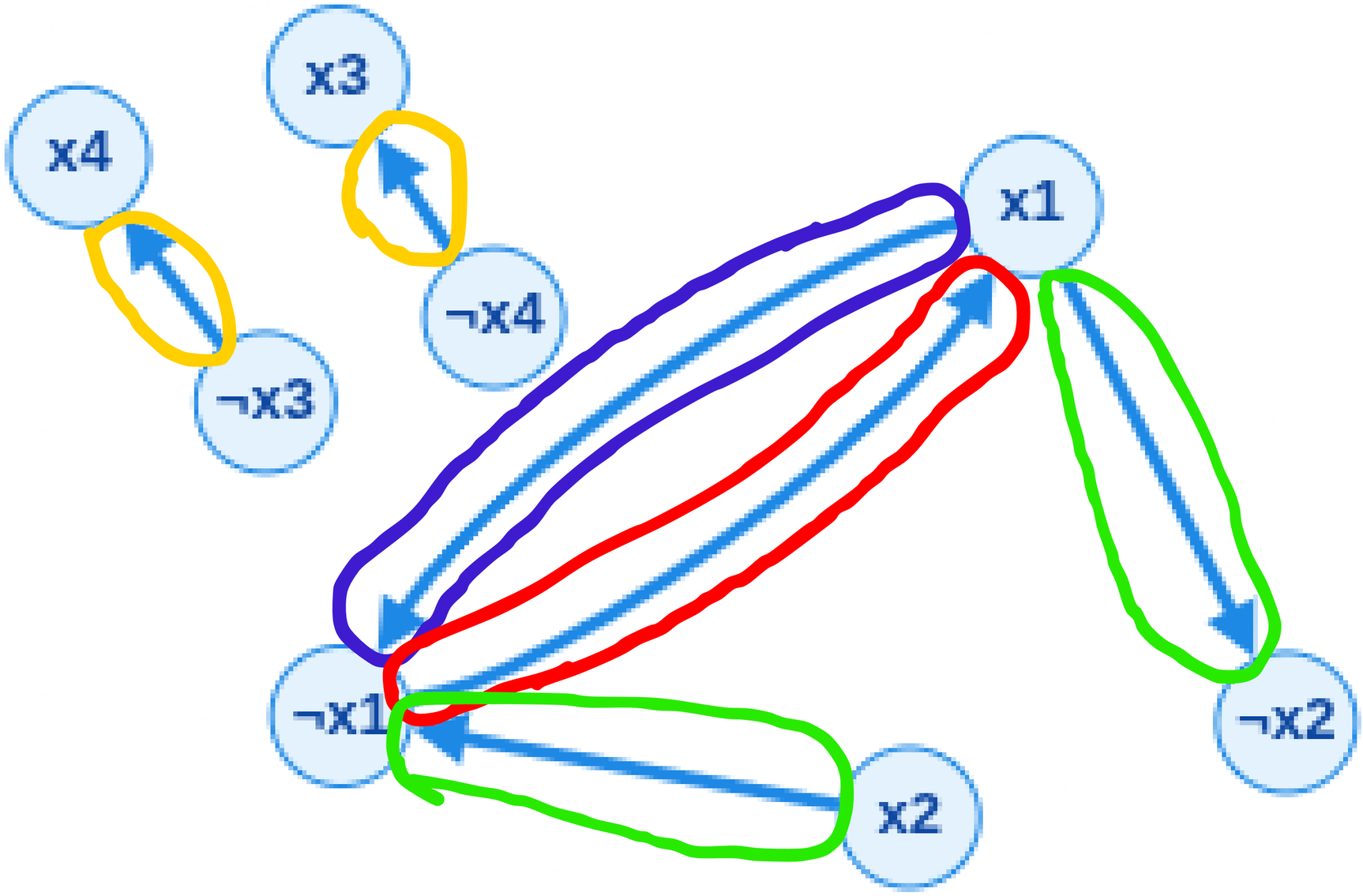


(x1 v ¬x2) (¬x3 v x4)

(x1 v x2) (¬x1 v x4)

(¬x1 v ¬x3)

c)



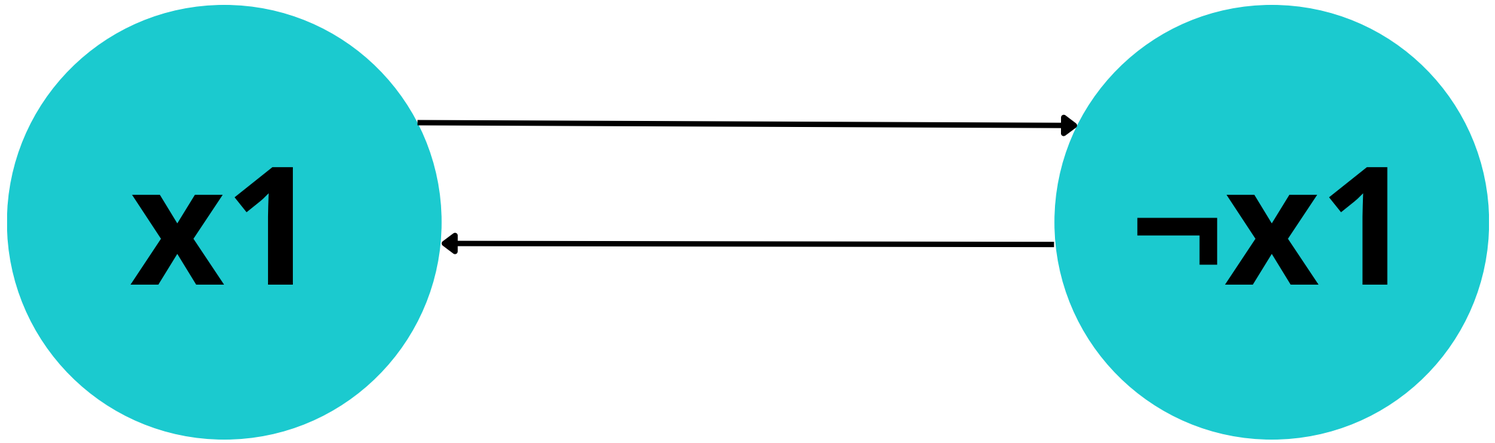
(x1 v x1)

(¬x1 v ¬x1)

(¬x1 v ¬x2)

(x3 v x4)

d)

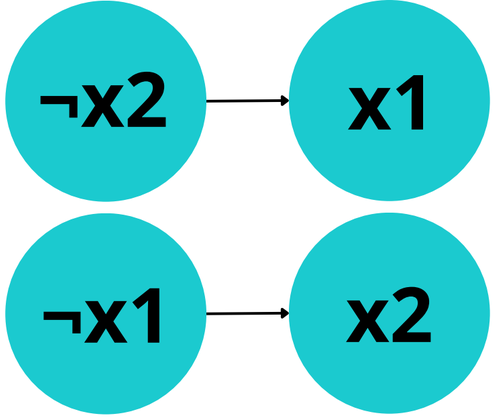


II

(x1 v x1) ∧ (¬x1 v ¬x1)

e)

G:



$\equiv$ (x1 v x2) $\equiv$ (¬x2 => x1)
(¬x1 => x2)

x_1	x_2	$\overline{x_2} \Rightarrow x_1$
0	0	0
0	1	1
1	0	**1**
1	1	**1**

0

0

f)

Require: Grafo de implicações $G = (V, E)$

1 $lista_CFC \leftarrow$ Tarjan(G)] O(|V|+|E|)
2 **for all** variável x_i **do**
3 **if** $CFC(x_i) = CFC(\overline{x_i})$ **then**
4 **return** Insatisfatível
5 **end if**
6 **end for**
7 Inicialize todos os literais como não atribuídos
8 **for all** CFC C em ordem decrescente de dependência **do**
9 **for all** literal $l \in C$ ainda não atribuído **do**
10 Atribua falso a l
11 Atribua verdadeiro a $\overline{l}$
12 **end for**
13 **end for**
14 **return** Atribuição encontrada

] O(|V|)

] O(|V|)

f)

Total: O(3*|V| + |E|)